

MASTER'S DEGREE IN DATA SCIENCE & AI
EVOLVE

MASTER'S THESIS

**Hospital Planning Support System Based On Real
SNS Data**

*A structural care-pressure index by pathology and autonomous community, buiilt and validated on
open public data from the INE and the Ministry of Health*

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Academic year

2025-2026

Madrid

Abstract

Hospitals in the Spanish National Health System (SNS; Ley 16/2003, 2003) manage the planning of their care capacity in a predominantly reactive way, without data-based tools that allow an objective comparison of where care pressure is concentrated across pathologies and territories. This Master's Thesis validates, file by file, eight public sources from the National Statistics Institute (INE, 2024) and the Ministry of Health (2025) and confirms a finding that determines the entire design of the project: none of them crosses diagnosis, territory and time simultaneously. Starting from this limitation - treated as a result, not as a footnote - the scope is narrowed to a single MVP: a structural care-pressure index by pathology and autonomous community for the year 2023, complemented by a descriptive block on the national trend 2014-2023.

Following the CRISP-DM methodology (Chapman et al., 2000), an integrated data layer is built (360 pathology × autonomous community combinations, with no null values, validated against the official discharge figure with a deviation of 0.11%) and a cross-sectional model is trained (Gradient Boosting Regressor; Friedman, 2001, with Ridge as a linear contrast alternative; Hoerl and Kennard, 1970) to explain the variable discharges per operating bed. Cross-validation, run under three different schemes - simple K-Fold, and GroupKFold by pathology and by autonomous community, the latter necessary because of the pseudoreplicated structure of the data (Hurlbert, 1984; Roberts et al., 2017) - reveals a central finding: the model's improvement over a simple baseline, which looked substantial (+8.6% in MAE) under naive validation, shrinks to a marginal +0.7% when the model is forced to generalise to territories never seen. The index separates pathologies robustly - dominated by their national clinical profile (death rate, transfer rate, emergency rate and average length of stay) - but does not support, with the available evidence, a robust territorial predictive capacity. This critical review, and not an optimistic starting figure, is documented as the main result of the evaluation.

The project is completed with the design of a prioritisation dashboard for medical management and bed management, explicitly communicated as a descriptive panel and not as a predictive or individual clinical tool, and with the conceptual - not implemented - design of two modules discarded for lack of a real source (30-day readmission risk and multi-year referral pressure), conditional on the future availability of sources with greater granularity. The work is delivered fully reproducible, with scripts and Google Colab notebooks run without errors on real data.

Keywords

National Health System, hospital planning, care pressure, CRISP-DM, health data science, open public data, cross-sectional, Gradient Boosting, group cross-validation, explainability (SHAP), reproducibility, data granularity limitations

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1. Business Understanding

1.1 Context: management in the SNS

The planning of care demand, the referral between levels of care and the pressure on hospital structures in the National Health System (SNS) are today managed in a predominantly reactive way (Ley 16/2003, 2003). Medical directorates and bed managers have historical and aggregated indicators, but not tools that allow them to systematically compare where real care pressure is concentrated - by pathology and by territory - in order to anticipate where to reinforce resources.

This reactive management is not only a problem of tools: it is also, as verified in the data validation phase of this TFM, a problem of the real availability of public information with the granularity needed to automate an operational response (for example, at weekly or service level). The project starts from that validation as an honest starting point, instead of assuming a granularity that the sources do not offer.

1.2 Problem addressed and who uses it

Problem: there is no single dashboard, based on real and public data, that allows an objective comparison of care pressure across autonomous communities and pathologies, or that places each territory against the national trend of the last decade.

Main users:

- **Medical directorate of a health area / network planning:** needs to identify in which pathology and territory combinations care pressure is structurally highest, in order to prioritise capacity reinforcement.
- **Bed management and resource allocation:** needs to contextualise current resourcing (beds, staff) against the volume of real activity in its community.

Decision supported: prioritisation of where to reinforce specialised capacity within a community or between communities, based on a comparable and explainable index, not on intuition or partial series.

1.3 Objectives

Business objective: to offer medical management and bed management a comparable care-pressure index by pathology and territory, supporting structural planning - not day-to-day operations.

Analytical objective: to build, from real public data from the INE (2024) and the Ministry of Health (2025), a structural care-pressure index (pathology × autonomous community/province, year 2023) by means of an interpretable cross-sectional model, complemented by a descriptive block on the national trend (2014-2023) that places this index in time.

It is important to distinguish both objectives: the business objective (helping to plan) is broader than what the analytical objective can demonstrate with the available data. The project is explicit about this difference at all times, instead of forcing the second to appear to fully cover the first.

1.4 MVP scope and justification

The initial scope of the project contemplated three modules: demand prediction (forecasting), referral pressure between levels of care, and 30-day readmission risk. After validating the eight real sources available (INE, 2024; Ministry of Health, 2025), it was found that none offers weekly granularity or a multi-year autonomous-community panel. What does exist, robustly and verifiably, is:

- A national annual series 2014-2023 (evolution of activity and resourcing at country level; Ministry of Health, 2025).
- A cross-sectional snapshot of the year 2023, with a rich crossing of diagnosis × province/autonomous community × discharge reason × average length of stay (INE, 2024).

Given this real limitation - not assumed, but verified file by file - the MVP is narrowed to a single module: the structural care-pressure index on the 2023 snapshot, with the 2014-2023 series as a trend and context block, not as a source of forecasting. The multi-year referral and readmission-risk modules remain documented as a conceptual, non-implemented design (Annex C), conditional on the future availability of sources with the granularity that does not exist publicly today (for example, iCMBD for readmission).

This scope decision is itself a result of the Data Understanding process (section 2), and is presented as such: the source limitation is not a footnote, but the finding that determines the design of the project.

1.5 Project success criteria

The project is considered useful if:

- The care-pressure index clearly discriminates between pathology × territory combinations with different structural pressure (not all values are similar).
- The model improves on a simple baseline (historical mean of the rate by pathology, without distinguishing autonomous community), showing that territory provides relevant information.
- The explanatory factors of the index are interpretable by a non-technical profile (medical management), not a black box (Lundberg and Lee, 2017).
- The project clearly distinguishes which conclusions the available real data support and which would fall outside scope for lack of a source (weekly forecasting, individual readmission).

The project would be considered failed, or its scope should be reviewed, if the index does not manage to separate territories/pathologies in a significant way (that is, if all variability is explained by noise or by the sample size in each cell), or if the only way

to sustain any of the discarded modules were to resort to synthetic data - something this TFM explicitly excludes as a methodological principle.

2. Data Understanding

2.1 Inventory of available sources

The project starts from eight public files from the INE (Hospital Morbidity Survey; INE, 2024) and the Ministry of Health (Statistics of Health Establishments with Inpatient Care, ESCRI; Ministry of Health, 2025), corresponding to the 2023 financial year except for two historical series 2014-2023:

File	Content	Geographic level	Available crossings
Alta_Diagnostico_Provincia.xlsx (INE, 2024)	Hospital discharges by main diagnosis	National, autonomous community (CCAA) and province	Diagnosis × sex × territory
Alta_Motivo_ingreso.xlsx (INE, 2024)	Discharges by discharge reason (cure, transfer, death, other)	National	Diagnosis × sex × reason
Alta_Urgencia_ingreso.xlsx (INE, 2024)	Discharges by admission type (ordinary / emergency)	National	Diagnosis × sex × urgency
Altas_Diagnostico_principal.xlsx (INE, 2024)	Discharges by age group	National	Diagnosis × sex × age
Estancia_media.xlsx (INE, 2024)	Average length of stay in days	National	Diagnosis × sex × age
Tablas_CCAA_2023.xlsx (Ministerio de Sanidad, 2025)	Hospitals, beds, staff, technology and diagnostic activity	CCAA	Multiple tables (1.1 to 7.3) by CCAA
Actividad_Evolucion_2014-2023.xlsx (Ministerio de Sanidad, 2025)	Historical series of hospital activity	National	Year (13 tables, 2014-2023)
Dotacion_Evolucion_2014-2023.xlsx (Ministerio de Sanidad, 2025)	Historical series of resourcing and staff	National	Year (10 tables, 2014-2023)

2.2 Validation of real granularity

This is the central finding of the data understanding phase, and the one that determines the scope of the project (see section 1.4). The following table summarises, for each source, which dimensions it really crosses:

Source	Time frequency	Geographic Level	Clinical level
Alta_Diagnostico_Provincia	Single snapshot (2023)	CCAA / province	Diagnosis (ICD-10 chapter and subchapter)
Alta_Motivo_ingreso	Single snapshot (2023)	National only	Diagnosis × discharge reason
Alta_Urgencia_ingreso	Single snapshot (2023)	National only	Diagnosis × urgency
Altas_Diagnostico_principal	Single snapshot (2023)	National only	Diagnosis × age
Estancia_media	Single snapshot (2023)	National only	Diagnosis × age
Tablas_CCAA_2023	Single snapshot (2023)	CCAA	Resourcing, staff, activity - not broken down by diagnosis
Actividad_Evolucion 2014-2023	Annual, 10 years	National	Activity aggregates (no diagnosis)
Dotacion_Evolucion 2014-2023	Annual, 10 years	National	Resourcing aggregates (no diagnosis)

The conclusion that already guided the index is confirmed with the real detail of each file: no source simultaneously crosses diagnosis + territory (CCAA/province) + time (several years). The only territory × diagnosis crossing available is a 2023 snapshot (Alta_Diagnostico_Provincia), and the only multi-year crossing (Actividad_Evolucion, Dotacion_Evolucion) is at aggregated national level, without diagnosis or territory. This rules out at the root any weekly forecasting or multi-year CCAA panel, and confirms that the structural index (2023 snapshot) and the national trend block (2014-2023 series) must be treated as two distinct analyses, not as a single table.

In addition, a further limitation not evident until the files were opened is confirmed: discharge reason, admission urgency and average length of stay exist only at national level, not crossed with province (INE, 2024). This means that the care-pressure index by CCAA/province (section 4) can only be built with the variables that do have that territorial crossing - fundamentally the volume of discharges by diagnosis (Alta_Diagnostico_Provincia) and the resourcing/staff from Tablas_CCAA_2023 (Ministry of Health, 2025) - while transfers, deaths, urgency and average length of stay are incorporated as national context by diagnosis, not as a territorial variable.

2.3 Structure and exploration of each file

All files share an INE-origin pattern: a descriptive header (title, units, year), a data table with the first column as diagnosis (coded according to ICD-10; World Health Organization [WHO], 2019, in hierarchical chapter/subchapter format) and subsequent columns as crossings (territory, sex, age, reason or urgency depending on the file).

- **Alta_Diagnostico_Provincia:** a single sheet (tabla-74901), with 63 territory columns (national, 17 CCAA and their provinces, plus Ceuta and Melilla) and diagnosis rows at two levels of detail (chapter, e.g. "0100 CERTAIN INFECTIOUS DISEASES...", and subchapter, e.g. "0101 Intestinal infectious diseases..."). It repeats the block for "Both sexes", "Men" and "Women" on the same sheet.
- **Alta_Motivo_ingreso / Alta_Urgencia_ingreso / Altas_Diagnostico_principal / Estancia_media:** one sheet each, with diagnosis structure in rows and the specific crossing in columns (reason, urgency or age), repeated by sex. No territory column.
- **Tablas_CCAA_2023:** a workbook with 24 sheets, one per numbered table (1.1 to 7.3) plus an index and two chart sheets. Each sheet has the CCAA in rows and that table's indicators in columns (hospitals, beds, staff, technology, diagnostic activity). There is no common diagnosis column that would allow these tables to be crossed with Alta_Diagnostico_Provincia at that level; crossing is only possible at aggregated CCAA level (Ministry of Health, 2025).
- **Actividad_Evolucion_2014-2023 / Dotacion_Evolucion_2014-2023:** each with an index and several sheets ("TABLE 3.1 3.2", "TABLE 1.1 1.2", etc.), with the years 2014-2023 as columns and the national aggregated indicators in rows (Ministry of Health, 2025).

2.4 Data quality

- **Hierarchical diagnosis coding:** each file mixes in the same column the chapter level (e.g. "0100 CERTAIN DISEASES...") and the subchapter level (e.g. "0101 Intestinal infectious diseases..."), identifiable by the numeric prefix (WHO, 2019). It is necessary to decide in data preparation which level to work at and filter out the other so as not to duplicate the total.
- **Statistical secrecy or non-significant values:** as is usual in INE tables with low counts per cell (INE, 2024), there are combinations marked as unavailable (..) that must be treated as missing, not as zero, so as not to bias the index downwards in small territories (Ceuta, Melilla, provinces with smaller populations).
- **Sheets with multiple tables or repeated headers:** several sheets (e.g. Tablas_CCAA_2023; Ministry of Health, 2025) include the full workbook title repeated on each tab before the real table, which requires skipping a variable number of header rows per sheet when parsing.

- **Non-normalised territory names between files:** for example, "Asturias, Principado de" in Alta_Diagnostico_Provincia versus "Asturias, Principado" in Tablas_CCAA_2023; a CCAA name normalisation dictionary will be necessary to be able to cross both sources reliably.

2.5 EDA if the national anual series 2014-2023

This section analyses the aggregated national evolution of hospital activity and bed resourcing over the last decade, based on the files Actividad_Evolucion_2014-2023.xlsx and Dotacion_Evolucion_2014-2023.xlsx (Ministry of Health, 2025). As indicated in section 2.2, this series offers only temporal breakdown, with no distinction by autonomous community or diagnosis, so its function in this work is to provide context and trend, and not to be the source of the structural index developed in section 4, which is built on the cross-sectional snapshot of the year 2023 (section 2.6).

Summary table of the 2014 - 2023 series

Year	Total discharges	Average stay (days)	Occupancy (%)	Emergencies	Public beds	Private beds	Total beds
2014	5.264.867	7,36	77,03	26.973.994	109.435	28.442	137.877
2015	5.285.563	7,36	77,29	28.053.453	110.150	27.814	137.964
2016	5.344.086	7,31	77,61	29.512.722	110.734	27.132	137.866
2017	5.336.131	7,29	76,94	30.030.788	111.598	26.983	138.581
2018	5.390.303	7,27	77,20	30.372.325	112.362	26.699	139.061
2019	5.383.760	7,23	76,84	31.342.724	112.264	26.597	138.861
2020	4.639.007	7,66	69,51	22.907.100	113.660	26.317	139.977
2021	4.942.588	7,46	71,94	27.980.842	114.195	26.202	140.397
2022	5.159.517	7,29	73,30	31.152.076	115.425	25.126	140.551
2023	5.301.878	7,11	73,97	33.375.738	117.482	22.207	139.689

Source: author's own elaboration based on Actividad_Evolucion_2014-2023.xlsx and Dotacion_Evolucion_2014-2023.xlsx (Ministry of Health, 2025).

Evolution of total hospital discharges

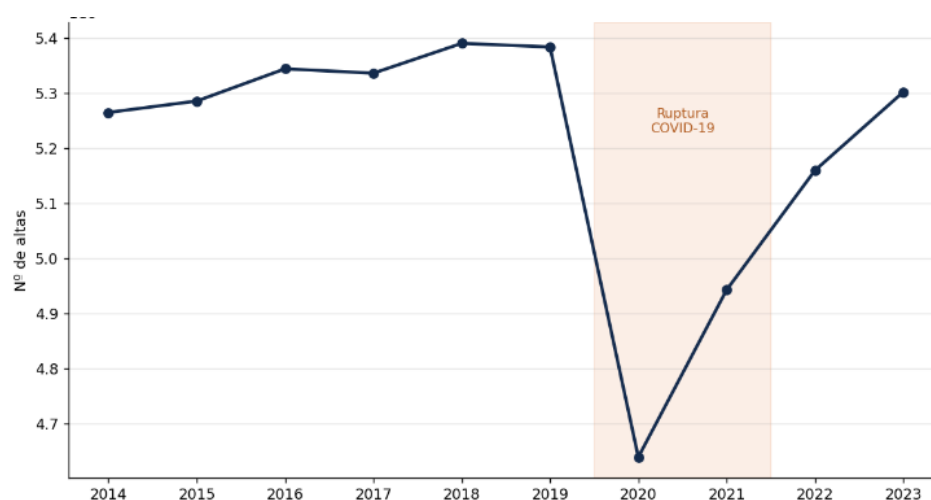


Figure 2.5.1. Evolution of total hospital discharges in Spain (2014-2023). Source: author's own elaboration based on Ministry of Health (2025). The figure is in the original PDF; the chart marks the 2020-2021 COVID-19 rupture

As seen in Figure 2.5.1, there is sustained growth between 2014 (5,264,867 discharges) and 2019 (5,383,760 discharges), equivalent to a cumulative increase of 2.3%. The rupture associated with the COVID-19 pandemic appears abruptly in 2020, with a fall to 4,639,007 discharges, a decrease of 13.8% with respect to the previous year and the largest year-on-year variation in the whole series (Ministry of Health, 2025). The subsequent recovery, between 2020 and 2023, is notable in relative terms (+14.3%), although incomplete: the volume of discharges in 2023 (5,301,878) remains 1.5% below the maximum reached in 2019.

Average stay and occupancy rate

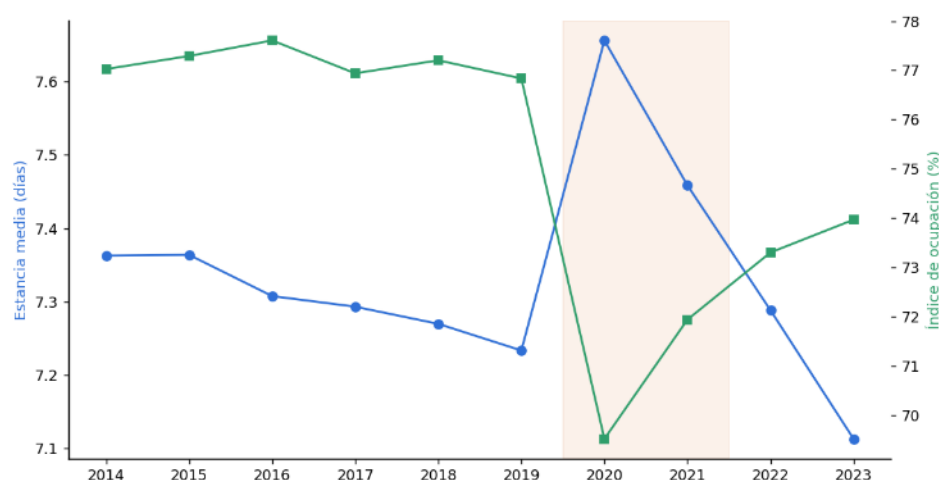


Figure 2.5.2. Evolution of average length of stay and hospital occupancy rate in Spain (2014-2023).

Source: author's own elaboration based on Ministry of Health (2025).

As Figure 2.5.2 shows, average length of stay presents a decreasing trend already before the pandemic, going from 7.36 days in 2014 to 7.23 days in 2019, which is compatible with a progressive improvement in bed-management efficiency. In 2020 there is a one-off spike to 7.66 days, the maximum value of the whole series, attributable to the combination of a lower number of total admissions and longer stays associated with severe COVID-19 cases. From 2021 the average stay resumes its previous downward trend, reaching its minimum value in 2023 (7.11 days).

The occupancy rate shows a behaviour opposite to what would be expected during the peak of the health crisis: instead of increasing, it falls from 76.8% in 2019 to 69.5% in 2020, as a result of the halting of a large part of scheduled activity. Recovery is progressive but, as with total discharges, the 2023 value (74.0%) still does not reach the level observed in 2019 (Ministry of Health, 2025).

Bed resourcing: Public versus private

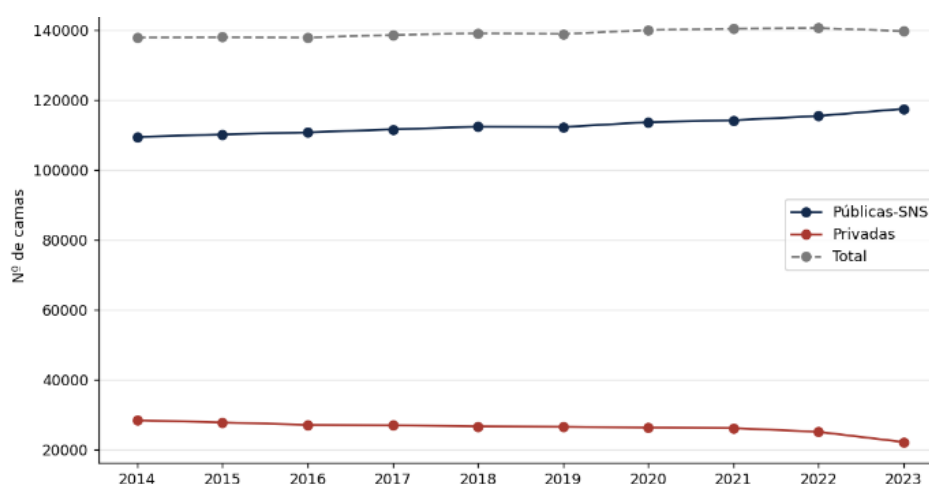


Figure 2.5.3. Evolution of hospital beds in operation, public and private, in Spain (2014-2023).
Source: author's own elaboration based on Ministry of Health (2025).

As seen in Figure 2.5.3, the national total of beds in operation remains practically stable over the decade, with a cumulative variation of 1.3% between 2014 and 2023. This aggregate stability hides, however, a relevant structural recomposition between the two subsectors: public beds of the National Health System increase by 7.3% (from 109,435 to 117,482), while private beds fall by 21.9% (from 28,442 to 22,207), with most of this decline concentrated in the last two years of the series (Ministry of Health, 2025). As a result, the relative weight of public resourcing over the total goes from 79.4% in 2014 to 84.1% in 2023.

Hospital emergencies attended

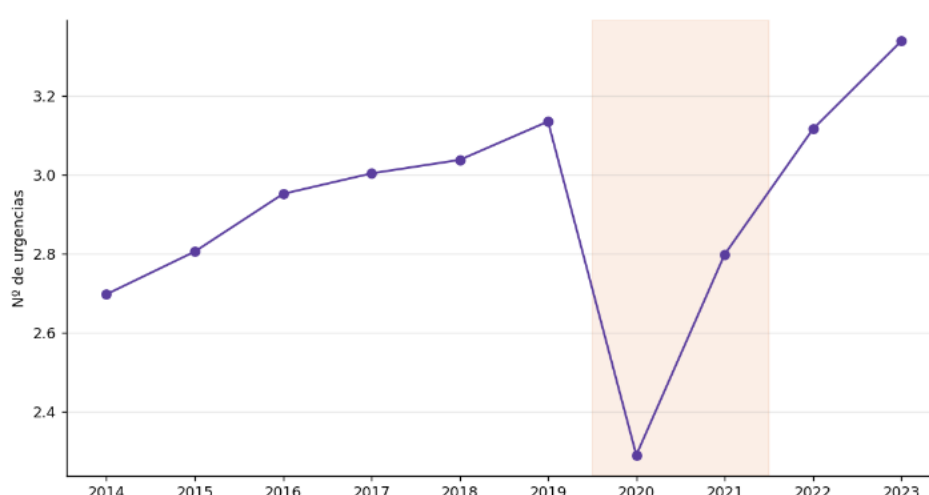


Figure 2.5.4. Evolution of hospital emergencies attended in Spain (2014-2023). Source: author's own elaboration based on Ministry of Health (2025).

As Figure 2.5.4 reflects, the evolution of hospital emergencies attended is the variable with the most pronounced structural growth of those analysed in this section. After the fall recorded in 2020 (-26.9% with respect to 2019, the sharpest of the whole series), the recovery not only reaches but clearly exceeds the pre-pandemic level, reaching in 2023 a historic maximum of 33,375,738 emergencies attended, 6.5% above the 2019 value (Ministry of Health, 2025). This behaviour contrasts with that of hospital discharges, which at the end of the series remain below their pre-pandemic level, and constitutes a first indication, at national level, of growing care pressure that motivates the development of the structural index proposed in this work.

Regarding the evolution of the national volume of hospital discharges and the magnitude of the 2020-2021 rupture, the data show moderate and sustained growth between 2014 and 2019 (+2.3%), followed by an abrupt fall in 2020 (-13.8% year-on-year, the sharpest in the series) and an incomplete recovery towards 2023, which stands 1.5% below the pre-pandemic maximum. It is therefore a clearly identifiable and quantifiable episode, although already overcome in terms of trend in the reference year used for the structural index (2023).

As for whether bed and staff resourcing has grown at the same pace as activity, the answer the available sources allow is partial. Bed resourcing has remained practically stable throughout the decade (+1.3% in total), well below the growth experienced by urgent demand (+6.5% with respect to 2019, with a historic maximum in 2023), which points to a growing gap between urgent care pressure and installed capacity. It is not possible, however, to extend this conclusion to staff resourcing, since the file *Dotacion_Evolucion_2014-2023.xlsx* does not include a historical series of staff, but only of beds, operating theatres and technological equipment (Ministry of Health, 2025). The only available staff data corresponds to the 2023 snapshot (*Tablas_CCAA_2023.xlsx*), which is incorporated as a structural variable of the index developed in section 4, and not as a time series.

2.6 EDA of the 2023 snapshot

This section analyses the cross-sectional snapshot of the 2023 financial year, the only year in which the available sources allow diagnosis to be crossed with territory (Alta_Diagnostico_Provincia.xlsx), as well as the discharge reason (Alta_Motivo_ingreso.xlsx) and the average stay by age group (Estancia_media.xlsx) at national level (INE, 2024). This block constitutes the empirical basis on which the structural care-pressure index developed in section 4 will be built.

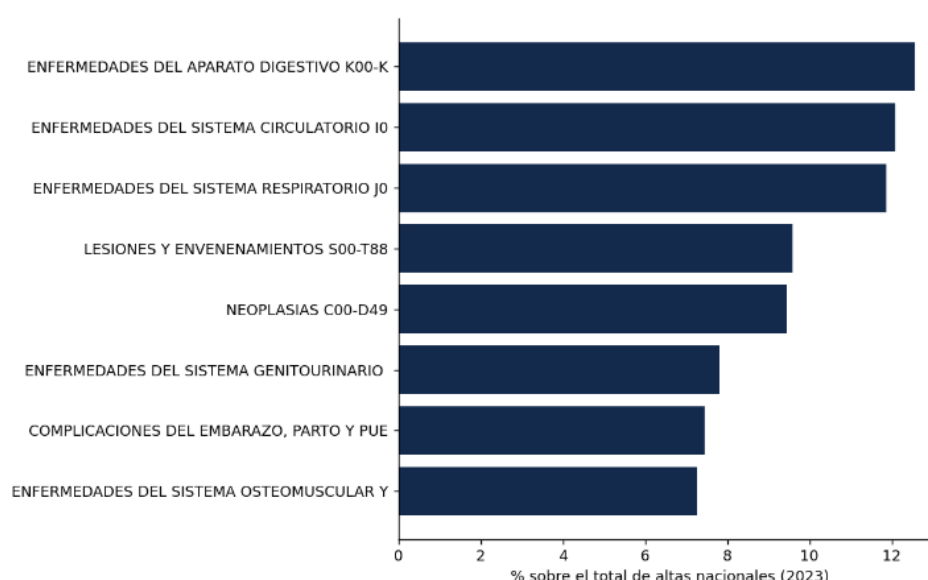


Figure 2.6.1. *Distribution of hospital discharges by diagnostic chapter at national level, Spain 2023. Source: author's own elaboration based on INE (2024).]*

As Figure 2.6.1 shows, out of a total of 4,866,745 hospital discharges recorded in Spain in 2023 (INE, 2024), three diagnostic chapters together account for slightly more than a third of the total volume: diseases of the digestive system (12.6%), diseases of the circulatory system (12.1%) and diseases of the respiratory system (11.9%). Next come injuries and poisonings (9.6%) and neoplasms (9.4%). This concentration in a small number of diagnostic chapters is consistent with usual clinical practice and is the starting point for assessing whether this distribution remains stable across autonomous communities or whether, on the contrary, there are differentiated territorial patterns.

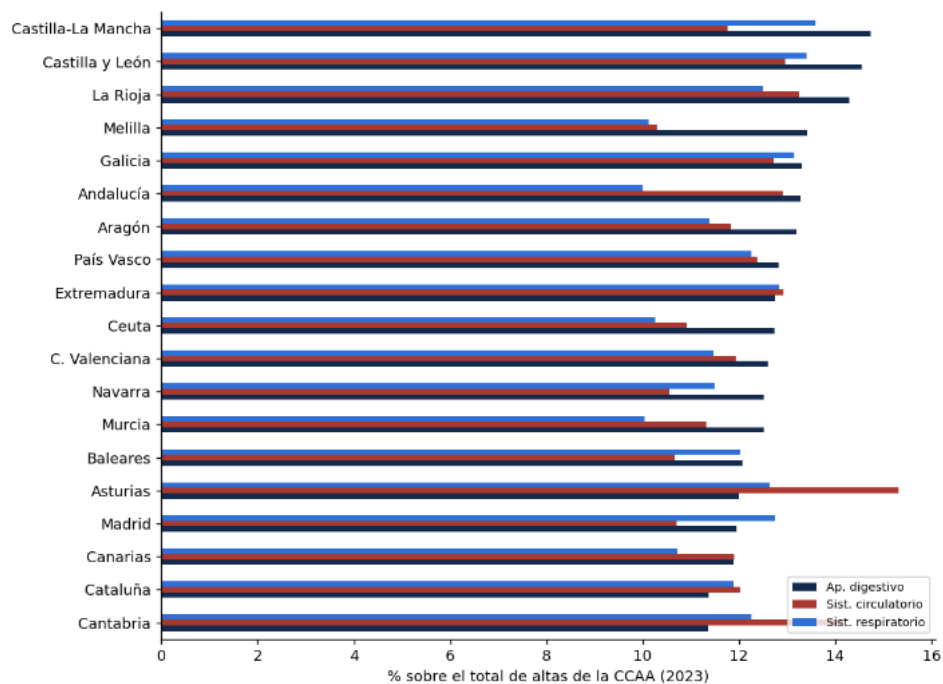


Figure 2.6.2. Weight of the three main diagnostic chapters over total discharges, by autonomous community, Spain 2023. Source: author's own elaboration based on INE (2024).

As seen in Figure 2.6.2, when comparing the relative weight of these three diagnostic chapters across autonomous communities, although the digestive system is the predominant chapter in most territories, there are relevant exceptions. In Asturias, Cantabria, the Canary Islands, Catalonia and Extremadura, the circulatory system displaces the digestive system as the main reason for discharge, reaching in Asturias a weight of 15.3% of the community's total discharges, the highest value observed for any chapter in any territory. In Madrid, for its part, it is the respiratory system that occupies first position (12.8%). At the other end, Castilla-La Mancha (14.7%) and Castilla y León (14.6%) show the highest relative concentration in the digestive system. This heterogeneity confirms that territory provides relevant information beyond the aggregated national distribution, which methodologically justifies building a care-pressure index broken down by autonomous community rather than limiting it to a national ranking of pathologies.

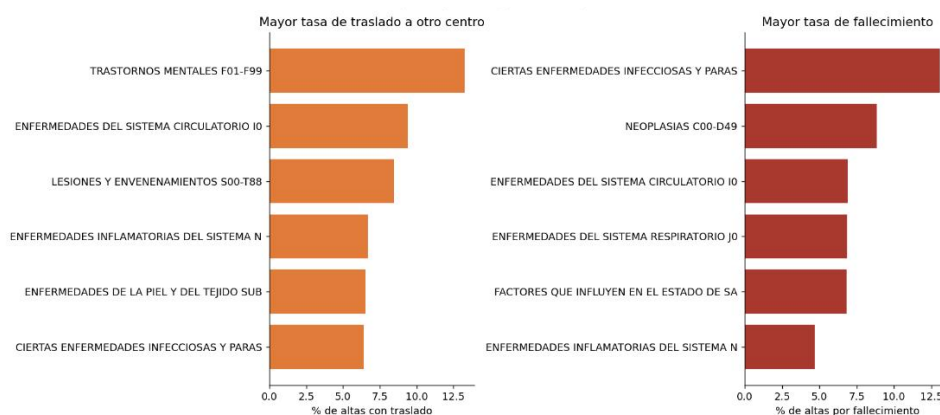


Figure 2.6.3. Transfer-to-another-centre and death rates by diagnostic chapter, Spain 2023. Source: author's own elaboration based on INE (2024). Left panel: highest transfer rates (mental disorders F01-F99, circulatory system, injuries and poisonings, inflammatory diseases, skin diseases, certain infectious and parasitic diseases). Right panel: highest death rates (certain infectious and parasitic diseases, neoplasms, circulatory system, respiratory system, factors influencing health status, inflammatory diseases).

As Figure 2.6.3 shows, at national level, 5.8% of hospital discharges in 2023 were due to transfer to another centre and 4.5% corresponded to death (INE, 2024). These overall rates, however, hide substantial differences between diagnostic chapters. The highest transfer rate corresponds to mental disorders, with 13.3% of discharges in that chapter resulting in transfer to another centre, more than double the overall rate and far above the second chapter, diseases of the circulatory system (9.4%). As for the death rate, the highest value is recorded in certain infectious and parasitic diseases (13.1%), followed by neoplasms (8.8%) and diseases of the circulatory system (6.9%). As pointed out in section 2.2, these rates are only available at national level, so it is not possible, with current sources, to determine whether these differences by diagnosis remain homogeneous across autonomous communities or whether there are, in addition, superimposed territorial differences.

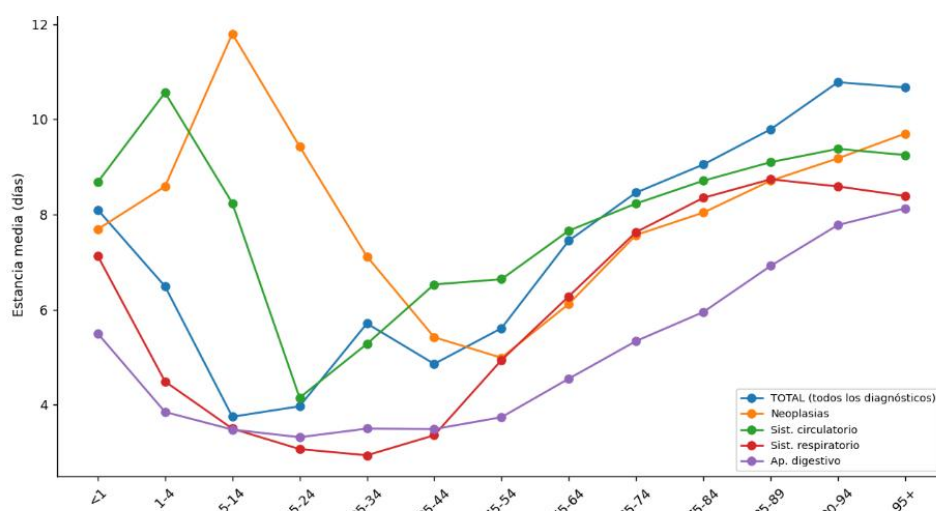


Figure 2.6.4. Average length of stay by age group and diagnostic chapter, Spain 2023 (excluding mental disorders, shown on a separate scale). Source: author's own elaboration based on INE (2024).

As observed in Figure 2.6.4, the relationship between average stay and age group presents, for most of the diagnostic chapters analysed, a U-shaped pattern: relatively high values in the youngest age groups (particularly in children under 1 year), a decrease to a minimum located between 15 and 34 years, and a progressive and sustained increase from that range up to the oldest groups. Thus, the average stay for all diagnoses goes from 8.10 days in children under 1 year to a minimum of 3.75 days in the 5-14 age group, to again exceed 9 days from age 65 and reach 10.78 days in the 90-94 group (INE, 2024). This pattern is repeated, with nuances, in chapters such as the digestive system or the respiratory system.

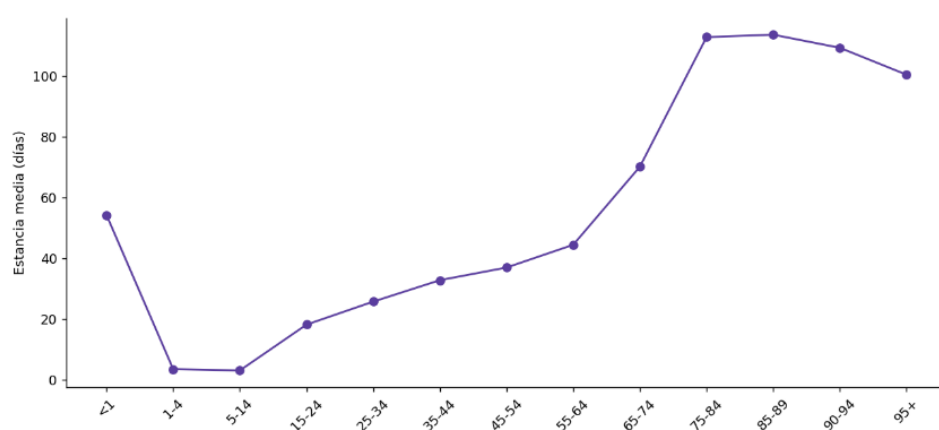


Figure 2.6.5. Average length of stay by age group, mental disorders chapter, Spain 2023. Source: author's own elaboration based on INE (2024).

As Figure 2.6.5 illustrates, the mental disorders chapter is an exception of far greater magnitude than the rest and is presented in a separate chart because of the difference in scale. Compared with single-digit average stays in the other chapters, mental disorders reach an average stay of 54.3 days in children under 1 year, fall to minimum values in childhood and adolescence (around 3 days between 1 and 14 years) and increase very sharply from adulthood onward, exceeding 100 days of average stay in the 75-94 age groups, with a maximum of 113.7 days in the 85-89 group (INE, 2024). This magnitude, between ten and fifteen times higher than that of the other diagnostic chapters in the oldest groups, is consistent with the nature of long-stay psychiatric admissions in the older population and constitutes one of the most relevant findings of this EDA block, although its clinical interpretation exceeds the analytical scope of this work.

In relation to the business questions posed for this block, the analysis confirms that, although the digestive system, the circulatory system and the respiratory system are systematically the three chapters of greatest weight across the national territory, the order among them varies systematically by autonomous community, with the circulatory system as the main diagnosis in five communities and the respiratory system as the main one in Madrid, which empirically justifies building the care-pressure index at autonomous-community level and not only at national level. As to which pathologies present the highest transfer or death rate, mental disorders stand

out clearly in transfers (13.3% versus a national average of 5.8%) and infectious and parasitic diseases lead the death rate (13.1% versus a national average of 4.5%), although this answer is limited to the national scope because of the absence of territorial breakdown in the corresponding source. Finally, regarding the relationship between average stay and age group by diagnosis, a general U-shaped pattern is confirmed in most chapters, with minima in young ages and maxima in the oldest groups, and mental disorders are identified as an exception of far greater magnitude than the rest of the diagnoses, with average stays exceeding 100 days in the oldest groups.

2.7 Conclusion of the phase

The file-by-file validation confirms and refines what was already anticipated for the index: the project can support, with 100% real data, a structural care-pressure index by diagnosis and CCAA/province for the year 2023 (built mainly on `Alta_Diagnostico_Provincia` crossed with the resourcing from `Tablas_CCAA_2023`; INE, 2024; Ministry of Health, 2025), and a descriptive national trend block 2014-2023 (on `Actividad_Evolucion` and `Dotacion_Evolucion`; Ministry of Health, 2025), treated as two separate analyses that cannot be combined with each other at detailed level.

The following are explicitly ruled out for lack of real granularity in the sources:

- Any weekly forecasting (no source has a frequency below annual).
- A CCAA × several years panel (the available historical series are national only).
- Crossing of discharge reason, urgency or average stay with territory (those three variables exist only at national level).
- 30-day readmission per patient (none of the eight sources contains an episode or patient identifier; see Annex C).

3. Data Preparation

This phase builds the gold layer on which the modelling of section 4 will rest: the structural care-pressure index by pathology and autonomous community, year 2023. The whole process described below is implemented in the script `tfm_pipeline_seccion3.py` (and in its notebook version for Google Colab), developed in Python with the pandas library (McKinney, 2010) and run end to end without errors on the 8 original files.

3.1 Selection of final tables and fields per dataset

Of the 8 sources validated in the Data Understanding phase (section 2), the gold layer of the structural index uses five:

File	What it provides	Level	Role in the gold layer
Alta_Diagnostico_Provincia.xlsx (INE, 2024)	Hospital discharges by ICD-10 chapter × autonomous community	CCAA/province, 2023 snapshot	Base variable of the index - only diagnosis × territory crossing available
Tablas_CCAA_2023.xlsx (tablas 1.1, 2.1, 4.1, 5.1, 5.2 y 6.4; Ministerio de Sanidad, 2025)	Hospitals, bed resourcing, staff, admission rates, hospitalisation indicators and emergencies	CCAA, 2023 snapshot	Territorial structural context
Alta_Motivo_ingreso.xlsx (INE, 2024)	Transfer to another centre and death by diagnosis	National	National context by pathology
Alta_Urgencia_ingreso.xlsx (INE, 2024)	% of emergency admission by diagnosis	National	National context by pathology
Estancia_media.xlsx (INE, 2024)	Average stay by diagnosis	National	National context by pathology

`Actividad_Evolucion_2014-2023.xlsx` and `Dotacion_Evolucion_2014-2023.xlsx` (Ministry of Health, 2025) are left out of this layer: they feed the national trend block, already worked on in the EDA of section 2, and cannot be combined with the 2023 cross-section without losing territorial or diagnosis detail.

Within `Alta_Diagnostico_Provincia`, `Alta_Motivo_ingreso`, `Alta_Urgencia_ingreso` and `Estancia_media`, only the ICD-10 chapter-level rows are selected (WHO, 2019) - a 4-digit code ending in 00 (e.g. 1100 = diseases of the digestive system), excluding both the grand total (0000) and the subchapters (e.g. 1101). This results in 20 diagnostic chapters, consistent with those already used in the EDA of section 2.

3.2 Cleaning

The data quality problems identified in section 2 are resolved as follows:

- **Multi-level headers and sheets with repeated title.** Each sheet of `Tablas_CCAA_2023.xlsx` requires skipping a different number of rows before reaching the real table (from row 3 in table 1.1 to row 7 in 5.1). This is solved by dynamically locating the header row of each sheet instead of assuming a fixed offset.
- **Non-normalised territory names between files.** `Alta_Diagnostico_Provincia` uses upper-case names ("ASTURIAS, PRINCIPADO DE"), whereas `Tablas_CCAA_2023` gives them in mixed case ("Asturias, Principado de"). `normalize_ccaa()` is implemented, which lower-cases, removes accents, commas and the letter ñ (replaced by n - without this step, País Vasco and Cataluña did not match the canonical dictionary) and does a prefix match against a list of 18 canonical territories.
- **Ceuta and Melilla.** In `Alta_Diagnostico_Provincia` they are two independent columns; in `Tablas_CCAA_2023` they already appear combined in a single row ("Ceuta y Melilla"). To be able to integrate both sources without losing that information, the two autonomous cities are aggregated into a single territorial unit throughout the dataset. This is an explicit design decision, not a hidden loss of granularity.
- **Statistical secrecy values (..).** They are treated as missing, never as zero (INE, 2024), so as not to bias the index downwards in smaller territories. In practice, none of the columns finally selected for the gold layer contains cells of this type after the selection of chapters and CCAA (see null validation in point 3.6).
- **snake_case.** All gold-layer fields are named in snake_case (`camas_funcionamiento_por1000`, `pct_fallecimiento_nacional`, etc.).

3.3 Integration

The integration combines, for each of the 18 territorial units (17 CCAA + Ceuta and Melilla), the discharge volumes of the 20 diagnostic chapters (INE, 2024) with six structural-context tables extracted from Tablas_CCAA_2023.xlsx (Ministry of Health, 2025):

Source table	Variables incorporated
1.1 Hospitals by ownership	hospitales_sns, hospitales_privados, hospitales_total
2.1 Bed resourcing	camas_instaladas_total, camas_funcionamiento_total, camas_funcionamiento_por1000, pct_camassns
4.1 Staff by employment link	personal_vinculado, personal_por1000
5.1 Admission rates	ingresos_total_ccaa, altas_total_ccaa, altas_por1000, estancias_por1000
5.2 Hospitalisation indicators	estancia_media_sns, ocupacion_sns, rotacion_sns, mortalidad_sns (and their private-subsector equivalents)
6.4 Emergency activity	urgencias_total, urgencias_por1000, presion_urgencias

For the variables of table 5.2, the indicators of the public-SNS subsector are used as the representative value for each CCAA: public resourcing concentrates between 69.8% and 100% of beds in operation depending on the territory (with an average close to 84%, already documented in the national trend EDA of section 2), so it is the most stable reference for comparing territories; the private subsector remains available in the gold layer as an additional variable, not as a substitute.

The result of the integration is a table of 360 rows (20 pathologies × 18 territories) with no null values, keyed by (ccaa, pathology).

As a first coherence check of the crossing, the correlation between camas_funcionamiento_por1000 and altas_por1000 at CCAA level is 0.72: territories with more beds per inhabitant tend to also record more discharges per inhabitant, an expected result that supports the quality of the integration and, at the same time, anticipates one of the risks that modelling must control (distinguishing resourcing from real demand).

3.4 Derived variables

From the three national sources by pathology, four rates are calculated, and from the diagnosis × CCAA crossing, two pressure ratios:

Variable	Calculation	Example (observed range)
tasa_traslado_nacional	transfers to another centre / total discharges, by pathology	Maximum in mental disorders (13.3%); minimum in complications of pregnancy, childbirth and the puerperium (1.0%)
pct_fallecimiento_nacional	deaths / total discharges, by pathology	Maximum in certain infectious and parasitic diseases (13.1%, see section 2); minimum in complications of pregnancy (0.004%)
pct_urgente_nacional	emergency admissions / total admissions, by pathology	Between 43.7% (eye diseases) and 80.6% (complications of pregnancy)
estancia_media_dias_nacional	average stay in days, by pathology	From 3.0 days (ear diseases) to 54.3 days (mental disorders, already analysed in the EDA of section 2)
altas_pct_sobre_ccaa	pathology discharges / total discharges of that CCAA	For the digestive system, between 11.4% (Cantabria/Catalonia) and 14.7% (Castilla-La Mancha)
altas_por1000hab_camás	pathology discharges / beds in operation of the CCAA	Proxy of structural pressure on installed resourcing

The resourcing ratios per 1,000 and 100,000 inhabitants (camas_funcionamiento_por1000, personal_por1000, urgencias_por1000, etc.) are not recalculated: they already come normalised by population at source in Tablas_CCAA_2023.xlsx (Ministry of Health, 2025), and are incorporated directly to avoid introducing our own population denominator that would not exactly match the one used by the INE/Ministry of Health.

As a first reading of the aggregated pressure variable (total discharges of the CCAA / beds in operation), the territory with the highest ratio is the Comunitat Valenciana (45.3 discharges per bed), and the lowest, the Canary Islands (27.0); that is, the most structurally pressured CCAA almost doubles ($\times 1.68$) the least pressured - the territorial variability that already justifies, at this stage, building an index broken down by CCAA in section 4.

3.5 Documented limitation

An express record is made of a real source limitation, already anticipated in section 2: tasa_traslado_nacional, pct_fallecimiento_nacional, pct_urgente_nacional and estancia_media_dias_nacional take the same value in all 18 territorial units for a given pathology, because none of their three sources (Alta_Motivo_ingreso,

Alta_Urgencia_ingreso, Estancia_media; INE, 2024) breaks down by CCAA or province.

This has been verified automatically in the pipeline: for mental disorders (code 0500), the variable `tasa_traslado_nacional` takes the value 0.1326 in the 18 rows corresponding to the 18 CCAA, with no variation. No territorial value has been interpolated, distributed or synthesised for these four variables. They are incorporated into the gold layer marked with the suffix `_nacional` precisely so that, in modelling and in the dashboard, they are not mistakenly read as a real difference between autonomous communities, but as pathology context applied equally to the whole territory.

3.6 Final gold layer

The pipeline generates three files:

- **gold_presion_asistencial_2023.csv** - primary key (anio, ccaa, patologia_cie10). 360 rows, 45 columns, 0 null values. This is the table that will feed the modelling of section 4.
- **gold_contexto_ccaa_2023.csv** - primary key (ccaa). 18 rows, 34 columns, 0 nulls. Complete structural context by territory, reusable outside the pathology crossing (for example, for the dashboard of section 6).
- **diccionario_datos_gold_2023.csv** - data dictionary generated automatically from the dataframe itself (field, type, nulls, example value), with the 45 columns of the main layer documented.

As a final quality control, the sum of discharges in the gold layer (4,861,330) is compared with the official national figure for 2023 (4,866,745, cited in the EDA of section 2; INE, 2024): the difference is 0.11%, attributable to statistical-secrecy residues in very low-volume combinations between diagnostic chapter and small territory, and is considered negligible for the purposes of the index.

3.7 What is discarded and why

Discarded element	Reason
ICD-10 subchapters (codes not ending in 00)	Work is done at chapter level so as not to duplicate the total when aggregating, and for consistency with the level of detail available in the rest of the sources
Years 2020-2021 in the national trend block	Pandemic structural rupture, already addressed and justified in the EDA of section 2
Crossing of transfer/death/urgency/average stay with CCAA or province	Does not exist in any of the 8 available public sources; it is documented as a real gap (point 3.5), not estimated or synthesised
Provinces (level below CCAA in Alta_Diagnostico_Provincia)	The rest of the structural-context sources (Tablas_CCAA_2023) do not go below CCAA level; descending to province would break the integration and generate very low-volume combinations
Sex as a crossing variable	The "Both sexes" block is used to keep the focus on the territorial index; it is documented as a future line of disaggregation (Annex)

3.8 Visual validation of the gold layer

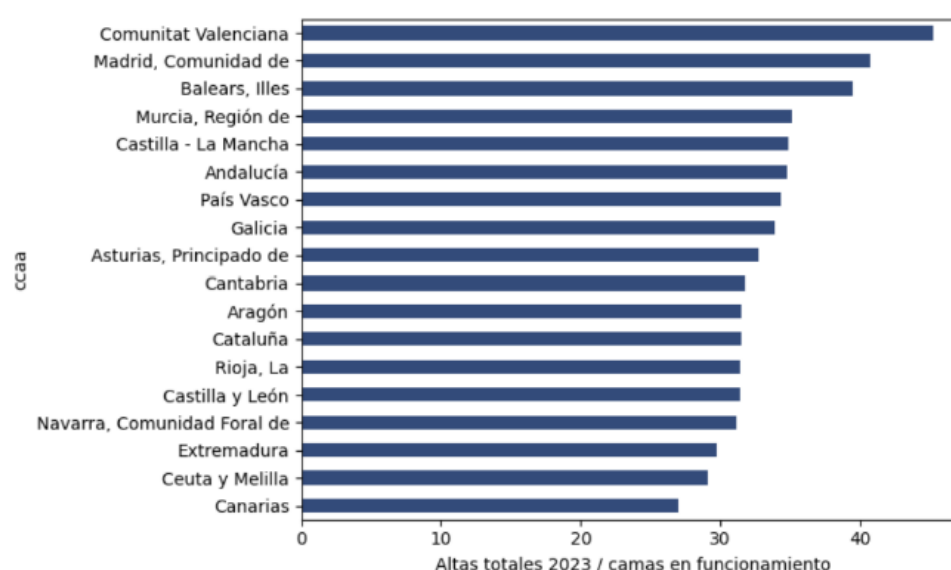


Figure 3.1. Aggregated structural pressure by autonomous community (total 2023 discharges / beds in operation), from the gold layer *gold_presion_asistencial_2023.csv* aggregated by territory. Source: author's own elaboration based on INE (2024) and Ministry of Health (2025).

As a final check of the integration - and a first preview of the type of reading the section 4 index will allow - Figure 3.1 aggregates the 360 rows of the gold layer by CCAA and confirms the territorial variability already pointed out in point 3.4: the Comunitat Valenciana (45.3 discharges per bed), Madrid (40.7) and the Balearic Islands (39.5) lead aggregated structural pressure, against the Canary Islands (27.0), Ceuta and

Melilla (29.1) and Extremadura (29.7) at the opposite end - a maximum-to-minimum ratio of 1.68. As it is a direct aggregation of the gold layer (and not an independent calculation), this figure also serves as verification that the diagnosis × CCAA × structural context integration has been carried out correctly for the 18 territorial units.

Closing methodological note: the gold layer built in this section supports, with 100% real data, a care-pressure index at pathology × CCAA level for 2023, with a block of national context by pathology clearly differentiated and marked as such. No variable has been estimated, interpolated or synthesised to complete crossings that public sources do not offer - the absence of those crossings is documented as a result, not concealed.

4. Modeling

4.1 Framing of the index

Target variable: `altas_por1000hab_camás` - discharges of a pathology in a CCAA divided by the beds in operation of that CCAA. It is the same variable already used as visual validation of the gold layer (Figure 3.1): the index does not start from an invented label, but from a real proxy directly computable from the gold layer, which allows the final "care-pressure index" to be the result of a model trained on that magnitude, with learned and justifiable weights, not a formula with weights set by one's own judgement.

Predictors, grouped into two levels (12 variables):

Level	Variables	Varies by
Pathology (national)	<code>tasa_traslado_nacional</code> , <code>pct_fallecimiento_nacional</code> , <code>pct_urgente_nacional</code> , <code>estancia_media_dias_nacional</code>	Only by pathology (constant across the 18 CCAA, see limitation documented in 3.5)
CCAA (structural context)	<code>personal_por1000</code> , <code>ocupacion_sns</code> , <code>rotacion_sns</code> , <code>mortalidad_sns</code> , <code>urgencias_por1000</code> , <code>presion_urgencias</code>	Only by CCAA (constant across the 20 pathologies)

`altas_pct_sobre_ccaa` is explicitly excluded as a predictor, even though it varies at segment level (pathology × CCAA) and might seem the most informative: it shares the numerator discharges with the target variable ($\text{altas_pct_sobre_ccaa} = \text{discharges} / \text{altas_total_ccaa}$ versus $\text{altas_por1000hab_camas} = \text{discharges} / \text{camas_funcionamiento_total}$). Although the denominators differ, both magnitudes are mechanically tied to the same real volume of discharges - including it would have been a partial information leak (Kaufman et al., 2012) disguised as a structural predictor, not an independent variable.

Why a cross-sectional model, not a time series: the diagnosis × territory crossing exists only as a single 2023 snapshot (section 2, "Validation of real granularity"); no public source offers it for several years.

Methodological warning for validation (developed in 5.1): the 360 rows of the gold layer are not 360 independent observations. The 4 pathology variables repeat across the 18 CCAA (pseudoreplication by pathology) and the 8 CCAA variables repeat across the 20 pathologies (pseudoreplication by CCAA; Hurlbert, 1984); the only component truly specific to each cell is discharges (and, by construction, the target variable). This does not invalidate the model, but it requires that the cross-validation of section 5 not be a simple random KFold - which would leak information from the same pathology or CCAA between training and test (Roberts et al., 2017) - and instead be complemented with GroupKFold by pathology and GroupKFold by CCAA.

4.2 Model alternatives and decisión rule

Los modelos se implementan con la librería scikit-learn (Pedregosa et al., 2011).

Alternative	Why it is proposed	Limitation / risk to control
Ridge regression (main candidate; Hoerl and Kennard, 1970)	Directly interpretable coefficients as the weight of each predictor in the index; L2 regularisation controls the expected collinearity among pathology variables (which covary: higher pct_fallecimiento_nacional tends to coincide with higher estancia_media_dias_nacional; Hastie et al., 2009)	Assumes a linear relationship between predictors and target; to be validated in the EDA of this phase before ruling out interactions
Gradient Boosting Regressor (contrast alternative; Friedman, 2001)	Allows capturing non-linear interactions (e.g., that severity matters more in CCAA with lower staffing) and would be used together with SHAP (Lundberg and Lee, 2017) to maintain explainability by segment	With 360 rows and a data structure with repeated blocks (see below), the risk of overfitting is higher than in Ridge; it is accepted only if it clearly improves explanatory capacity in the cross-validation of section 5

Decision rule: Ridge is prioritised as the model unless Gradient Boosting provides a substantial and verifiable improvement. The results of section 5 show that Ridge substantially worsens the baseline under all three validation schemes (non-linear relationship), whereas GBR improves it - although, as detailed in 5.2/5.3, that improvement depends critically on which validation scheme is used, and is much more modest than the simplest validation suggests.

4.3 National trend block 2014-2023

With 10 annual observations aggregated at national level (without diagnosis or CCAA; Ministry of Health, 2025), this block is kept explicitly descriptive, not predictive:

- **What can be said:** the magnitude and direction of the 2020-2021 rupture, the cumulative pre-pandemic growth rate (2014-2019, +2.3%) and post-pandemic (2020-2023, +14.3% relative but incomplete), the decoupling between the growth of emergencies (+6.5% over 2019, historic maximum) and the practical stability of bed resourcing (+1.3% over the decade), and the recomposition of the public/private weight in resourcing - see Figures 2.5.1 to 2.5.4.
- **What cannot be said with guarantees:** a projection of future values. Ten annual points do not offer the resolution needed to fit a time-series model with statistical validity (neither to decompose seasonality - there is none, it is an annual series - nor for a reliable prediction interval 1-2 years ahead). Fitting a simple linear regression on 10 points and projecting it would be technically possible but methodologically weak: the confidence interval would be artificially

narrow and would not reflect the real uncertainty of extrapolating a decade-long trend with a single known rupture point (the pandemic) and, potentially, other unobserved ones. This block provides trend context, not projection, and does not compete with the index of 4.1 nor aim to replace it.

4.4 Discarded variables and reason

Discarded variable / element	Reason
camas_funcionamiento_por1000, altas_por1000 as predictors	Circularity: both are structurally tied to the numerator/denominator of the target variable (altas_por1000hab_cammas); including them would artificially inflate the model's explanatory capacity without adding new information (Kaufman et al., 2012)
Province as the model's aggregation level	No structural-context source (Tablas_CCAA_2023) goes below CCAA level; modelling at province level would break the integration already documented in section 3
Sex as a model variable	Kept out of the main index for the same reason documented in data preparation (section 3.7); remains as a future line of disaggregation
Any segment-level forecasting (pathology × CCAA × year)	There is no multi-year panel with that granularity (see section 2.2); ruled out at the root, not simplified with assumptions
Readmission or multi-year referral score as part of this model	They depended on multi-year sources not available at this granularity (iCMBD for readmission, daily occupancy for referral); they remain in Annex C as a conceptual, non-implemented design

4. 5 Explainability

For the Ridge model, each standardised predictor provides a coefficient that represents its relative weight in the index (Hastie et al., 2009); since it is a linear model, the contribution of each variable to the specific value of a segment (pathology × CCAA) is exactly decomposed as the product coefficient × standardised value of the variable in that segment - an approximation method such as SHAP is not needed to obtain it, although the result is communicated in the dashboard with the same "main factors" format already used in the front-end design of previous deliverables.

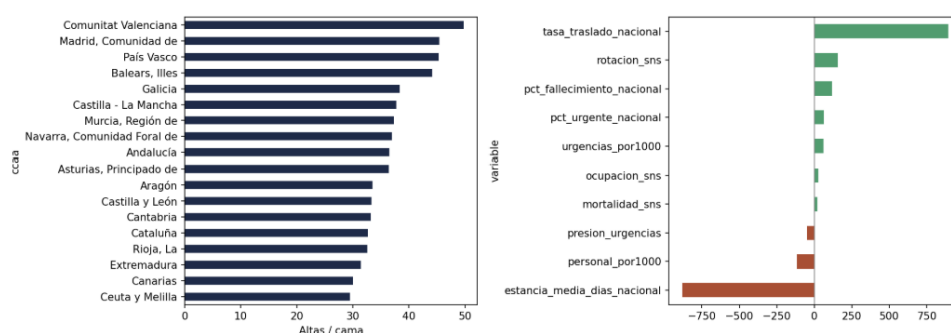


Figure 4.1: index modelling. Source: author's own elaboration. Left panel: aggregated structural pressure by CCAA (total 2023 discharges / beds in operation). Right panel: coefficients of the Ridge model (standardised predictors) - the largest positive coefficient is *tasa_traslado_nacional* and the largest negative is *estancia_media_dias_nacional*.

If the Gradient Boosting Regressor is validated as the preferable alternative in section 5, explainability rests on SHAP (Lundberg and Lee, 2017; Lundberg et al., 2020), which in tree models does not offer an exact decomposition but does offer a consistent and comparable attribution across segments.

In both cases, the result that reaches the dashboard per segment is the same format: the score rescaled from 0 to 100 and its 2-3 highest-contribution factors, expressed in non-technical language (e.g. "average stay far above the national median" instead of "coefficient of *estancia_media_dias_nacional* = 0.34").

5. Evaluation

5.1 Validation Strategy

Three cross-validation schemes are run (5 folds each), not just one, because they answer different questions and the methodological warning of 4.1 requires not settling for the most optimistic one. Standard cross-validation assumes independent observations; when data show hierarchical or grouped structure, it must be performed by blocks to avoid biased estimates of predictive capacity (Roberts et al., 2017):

1. **Simple random K-Fold** - initial reference; it overestimates performance because, within the same fold, rows of the same pathology or the same CCAA appear simultaneously in training and test (Kaufman et al., 2012; Roberts et al., 2017).
2. **GroupKFold by pathology** - leaves entire pathologies out; measures whether the model generalises to pathologies never seen.
3. **GroupKFold by CCAA** - leaves entire autonomous communities out; measures whether territorial structural context generalises to territories never seen. This is the validation that directly answers the central question of the project: does territory provide relevant information?

No temporal split is used: it is a cross-section with no time dimension in the pathology × CCAA crossing (section 4.1). The validation schemes have been implemented with scikit-learn (Pedregosa et al., 2011).

5.2 Metrics and baseline

Baseline: for each fold and each scheme, the mean of `altas_por1000hab_camias` for each pathology is computed using only the training rows of that fold, and that mean is used as the prediction - "the historical mean by pathology, without distinguishing CCAA" required by the project's success criterion.

Results of the three cross-validation schemes (mean MAE, 5 folds):

Scheme	Baseline MAE	GBR MAE	GBR improvement	Ridge MAE	Ridge improvement
Simple K-Fold	0.2659	0.2431	+8.6%	1.1655	-338%
GroupKFold by pathology	1.3532	1.2807	+5.4% (negative R ² in both)	2.2589	-67%
GroupKFold by CCAA	0.2801	0.2782	+0.7%	1.1136	-298%

5.3 Acceptance criterion

This is the central methodological correction of the evaluation: simple K-Fold overestimates the contribution of territory (+8.6%) because it leaks information from the same CCAA between training and test (Roberts et al., 2017). When the model is forced to generalise to territories never seen (GroupKFold by CCAA, the validation that really answers the project's question), the improvement of GBR over the baseline falls to +0.7% - practically indistinguishable from the baseline.

The project's success criterion requires the model to improve on the baseline, showing that territory provides relevant information, without setting a strict numerical threshold. Under the correct validation, that improvement is real but marginal (+0.7%), well below what simple K-Fold suggested. Ridge, under all three schemes, substantially worsens the baseline. The model is accepted with this explicit caveat: the honest finding - not the initial optimistic figure from simple K-Fold - becomes the central result of the critical review (5.4). Had none of the three schemes shown any improvement, the documented result would have been the descriptive panel of Figure 3.1, with no added predictive component.

5.4 Critical review

With the three validation schemes in hand, the answer is qualified with respect to a first optimistic reading:

- The dominance of the clinical profile of the pathology (mental disorders with longer stay and transfer, already documented in the EDA of section 2, Figure 2.6.5) is the most solid finding, and at the same time the most limited: the baseline by pathology explains an R^2 of 0.90-0.91 under simple K-Fold, but collapses to a negative R^2 when generalising to pathologies never seen (GroupKFold by pathology). That is, the model's "knowledge" about a pathology depends on having seen it in some CCAA during training, not on a transferable pattern derived from its 4 national clinical variables.
- The territorial contribution, which under simple K-Fold seemed real (+8.6%, with `personal_por1000` and `rotacion_sns` as dominant factors), is reduced to +0.7% under GroupKFold by CCAA. The honest conclusion is that, with the data and granularity available, there is no solid evidence that the structural context of a CCAA allows its care pressure to be predicted in unseen territories. The index retains its usefulness as a descriptive and ranking panel among the 18 CCAA and 20 pathologies already known (Figure 3.1), but must not be presented as a predictive model with demonstrated territorial generalisation capacity.
- This is exactly the critical review demanded by the project's own success criterion: the index separates pathologies clearly and consistently, but does not separate territories with robust evidence once the validation bias is corrected - a more modest, and more honest, result than the first estimate with simple K-Fold.

5.5 Risks and alternatives

Risk	Description	Treatment
Overestimation due to leakage between folds	A simple random K-Fold leaks information from the same pathology/CCAA between training and test (Kaufman et al., 2012)	Corrected with GroupKFold by pathology and by CCAA (5.1-5.3; Roberts et al., 2017)
Lack of generalisation to new pathologies	Negative R^2 in GroupKFold by pathology, for both the baseline and GBR	Documented as a limitation; the index is used as a ranking of already-known pathologies, not as a classifier of new pathologies
Lack of generalisation to new territories	Almost null improvement of GBR (+0.7%) in GroupKFold by CCAA	Documented as the central limitation of the project; the index is not presented as a validated territorial predictive model
Confusion between clinical severity and network capacity	A low personal_por1000 could reflect either lower real resourcing or a different case mix	The model does not distinguish causality, only association - explicit interpretation limit, not resolved with the available data
Low volume in infrequent combinations	Small cells (Ceuta and Melilla, La Rioja) with low discharge counts can inflate the variance of the target variable	Visible in the standard deviation of MAE across folds; not corrected artificially

5.6 Explainability

SHAP is used (TreeExplainer; Lundberg and Lee, 2017; Lundberg et al., 2020) on the final GBR model (trained on 100% of the data, once the model choice is fixed): in tree models it does not offer an exact decomposition as in Ridge (coefficient $\times$ standardised value), but it does offer a consistent and comparable attribution across segments.

Mean |SHAP| importance (global, highest to lowest):

Variable	SHAP importance
pct_fallecimiento_nacional	0,7347
tasa_traslado_nacional	0,3352
pct_urgente_nacional	0,2079
estancia_media_dias_nacional	0,1993
rotacion_sns	0,0793
personal_por1000	0,0563
mortalidad_sns	0,0442
ocupacion_sns	0,0423
pct_camas_sns	0,0180
presion_urgencias	0,0104
estancia_media_sns	0,0100
pct_urgencias_ingresadas	0,0077

For the dashboard of section 6, each segment (pathology $\times$ CCAA) receives a score rescaled from 0 to 100 and its 2 main factors (highest |SHAP| in that row), in non-technical language. Real example from the extremes of the index:

- **Top:** Comunitat Valenciana + diseases of the circulatory system $\rightarrow$ score 100.0 (factors: pct_fallecimiento_nacional, personal_por1000).
- **Bottom:** Ceuta and Melilla + congenital anomalies $\rightarrow$ score 0.0 (factors: tasa_traslado_nacional, pct_fallecimiento_nacional).

Reading, with the same honesty as 5.4: the 4 national clinical-profile variables concentrate most of the SHAP importance; among the structural ones, rotacion_sns and personal_por1000 are the ones that weigh most, consistent with the 5.4%-8.6% territorial contribution seen in simple K-Fold, and with its practical disappearance (+0.7%) when generalising to new territories. The "main factors" that reach the dashboard are therefore mostly pathology factors, not territory factors - and this must also be communicated in section 6, so as not to suggest to the end user a territorial predictive capacity that the validation does not support.

6. Deployment

6.1 Dashboard

The dashboard is organised in two blocks, consistent with the separation already established between the 2023 snapshot and the historical series (sections 2 and 3):

Block 1 - Care-pressure index (2023). It directly consumes `indice_presion_dashboard.csv` (generated in `tfm_04_modelado.ipynb`): 360 rows with `ccaa`, `patologia_nombre`, `altas_por1000hab_camas`, `score_presion_0_100` and `factores_principales`. It is presented as:

- A ranking/map by CCAA, aggregating the score by territory (same reading as Figure 3.1: Comunitat Valenciana, Madrid and the Balearic Islands at the top; the Canary Islands, Ceuta and Melilla and Extremadura at the bottom).
- A ranking by pathology, showing which pathology × CCAA combinations concentrate the highest score (the real example already calculated: Comunitat Valenciana + diseases of the circulatory system, score 100.0).
- For each selected segment, its 2 main factors in non-technical language (e.g. "it is influenced above all by the death rate typical of this pathology at national level, and to a lesser extent by the healthcare staff per 1,000 inhabitants of this community").

Block 2 - National trend (2014-2023). It reuses Figures 2.5.1 to 2.5.4 of the EDA of section 2 (total discharges, average stay and occupancy, bed resourcing, emergencies; Ministry of Health, 2025), with the 2020-2021 rupture visually flagged. It is presented as a context panel, not filterable by CCAA or by pathology, because the source does not allow it (section 2.2).

Explicit design restriction, derived from section 5.4: the dashboard must not present the score as a validated prediction of future pressure by territory. The evaluation (section 5) showed that the territorial contribution to the index, although real, is modest and does not generalise robustly to CCAA never seen (+0.7% improvement under GroupKFold by CCAA; Roberts et al., 2017). The dashboard communicates the score for what it is: a descriptive ranking of 2023, built with a model whose weights are mostly determined by the clinical profile of the pathology, not by territory.

6.2 Target audience and how the result is communicated

Target audience, inherited from section 1 (Business Understanding):

- Medical directorate of a health area / network planning.
- Bed management and resource allocation.

How the result is communicated:

- The 0-to-100 score is always presented as a prioritisation index for structural planning, never as an operational alert or a decision about a specific patient - consistent with the business objective defined from section 1 ("support for structural planning, not day-to-day operations").
- The "main factors" are communicated in business language (e.g. "typical mortality of the pathology", "staffing of the community"), never as technical variable names or coefficients (Lundberg and Lee, 2017).
- A visible warning - not a footnote - is included that most of the score reflects the clinical profile of the pathology, and that the territorial component, although real, is the one with the least weight - thus preventing a non-technical user from interpreting the ranking between CCAA as a difference in management capacity more marked than the data support.
- No dashboard screen allows filtering or querying by individual patient: the entire gold layer is aggregated at pathology × CCAA level, and so it is kept in the final product.

6.3 Maintenance and updating

The INE (2024) and the Ministry of Health (2025) publish these eight sources annually. The pipeline of section 3 (`tfm_pipeline_seccion3.py`) is designed to be re-run on the 2024 publication without structural changes, except for two documented points of attention:

- The header-row offsets (`header_row_idx`) of `Tablas_CCAA_2023.xlsx` could vary from one year to another if the INE restructures a sheet; it is recommended to revalidate the dynamic header location (already implemented in point 3.2) before accepting the run.
- Territory names should continue to match `normalize_ccaa()`, but it is advisable to revalidate the count of 18 territorial units after each update (the function itself already includes an assert for this).

A medium-term maintenance opportunity, not available today: each additional annual publication of `Alta_Diagnostico_Provincia` is, in itself, a new pathology × CCAA cross-sectional snapshot. By accumulating several years of this same source (2023, 2024, 2025...), it would be possible to organically build the CCAA × several years panel that does not exist today (section 2.2) - without depending on new sources, simply by accumulating successive runs of this same pipeline. It is a natural path of evolution of the project, documented here and taken up again in 6.4 and 7.3.

6.4 Future extensions

Any extension is conditional on the availability of additional real sources, never on filling gaps with synthetic data - a methodological principle of the project since section 1:

Extension	Condition to activate it
30-day readmission risk score (Annex C, M3)	Availability of iCMBD (or equivalent) as a real aggregated indicator; as long as the readmission rate does not come from a real source, it is not trained (see Annex C)
Multi-year referral pressure score (Annex C, M2)	A file with a time series of transfers by CCAA, non-existent today at that granularity
CCAA × several years panel for the pressure index	Accumulation of successive annual publications of Alta_Diagnostico_Provincia (see 6.3) - the only extension on this list that does not depend on a new source, only on the passage of time
Breakdown by sex	Already available in the current sources ("Men"/"Women" blocks discarded in 3.2/3.7 for scope, not for lack of data)
Explainability by segment with alternative models	Contrast of SHAP-GBR against Ridge coefficients already calculated (section 4.5); extendable if a source with a greater volume of observations is incorporated, reducing the overfitting risk noted in 5.5

7. Conclusions and limitations

7.1 What the Project demonstrates and what it does not

What it demonstrates:

- That it is possible to build, with 100% real and public data from the INE (2024) and the Ministry of Health (2025), an integrated and validated gold layer (360 rows, 0 nulls, deviation of 0.11% against the official discharge figure) that crosses diagnosis and territory for the year 2023 - something none of the eight sources offers separately.
- That the resulting care-pressure index separates pathologies clearly and consistently (the national clinical profile explains an R^2 of ~ 0.91 over the target variable).
- That, once the validation bias is corrected (GroupKFold by CCAA; Roberts et al., 2017), there is a real but modest territorial contribution (+0.7% MAE improvement), concentrated in `personal_por1000` and `rotacion_sns`.
- That a model of this type can be documented with the same methodology (CRISP-DM; Chapman et al., 2000), reproducibility (script + notebook run without errors) and critical honesty that a real production environment would demand.

What it does NOT demonstrate:

- It does not demonstrate robust territorial generalisation capacity: under the correct validation, the model barely improves on predicting with the historical mean of each pathology.
- It does not demonstrate forecasting capacity: the national trend block 2014-2023 is descriptive, with only 10 annual points, insufficient to project with guarantees (section 4.3).
- It neither demonstrates nor evaluates individual readmission risk or multi-year referral: they remain a conceptual, non-implemented design (Annex C), conditional on sources that do not exist publicly today.
- It does not establish causal relationships: the association between `personal_por1000` and care pressure does not imply that increasing staff would reduce pressure - the model does not distinguish causality from association (section 5.5).

7.2 Source limitations as the central result of the TFM

The granularity limitation of the public sources - none crosses diagnosis + territory + time simultaneously - is not a footnote of this work: it is the finding that determined its design from the Data Understanding phase (section 2) and that is sustained, verified, through to the final evaluation (section 5). The project documents this limitation explicitly in each phase in which it is relevant:

- In scope selection (section 1.4): a single module (2023 structural index) instead of the three modules initially contemplated.
- In data preparation (section 3.5): discharge reason, urgency and average stay only at national level.
- In modelling (section 4.1): exclusion of variables due to information leakage derived from the structure of the gold layer itself (Kaufman et al., 2012).
- In evaluation (sections 5.3-5.4): the correction of validation bias that revealed that the territorial signal is weaker than a first approximation suggested.

Treating the source limitation as a result, and not as an excuse, is what makes the conclusions of this TFM defensible: every claim is bounded exactly to what the available real data can support.

7.3 Future lines

- Organic temporal accumulation (see 6.3): repeat the pipeline run in successive years to build, over time, the CCAA × several years panel that does not exist today - the only future line that does not depend on a new source.
- Incorporation of iCMBD (or equivalent) to activate the conceptual design of 30-day readmission risk (Annex C, M3), maintaining the methodological correction already documented (real patient identifier, validation by groups, not by episodio_id; Roberts et al., 2017).
- A transfers source with territorial breakdown, to activate the conceptual design of referral pressure (Annex C, M2), today limited by the absence of that breakdown in Alta_Motivo_ingreso (INE, 2024).
- Breakdown by sex, already available in the current sources but discarded for scope (section 3.7), as a low-cost extension on the same pipeline.
- A greater volume of observations before reconsidering models with more parameters (for example, pathology or CCAA dummies, discarded in 4.4 due to overfitting risk on 360 rows): each additional accumulated year (see line 1) reduces that risk and widens the available modelling alternatives.

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Annexes

A. Complete data dictionary. B. Detail of the 8 original sources and their technical sheet (INE/Ministry of Health). C. Conceptual design not implemented (multi-year referral, readmission risk) - inherited from Deliverables 3-4, conditional on new sources.

Annex A. Complete data dictionary

This annex documents the 45 variables of the main gold layer, gold_presion_asistencial_2023.csv (360 rows = 20 pathologies × 18 territories, primary key (anio, ccaa, patologia_cie10), 0 null values), prepared from INE (2024) and Ministry of Health (2025). The 34 structural-context variables (blocks B to D) coincide with those of gold_contexto_ccaa_2023.csv (18 rows, primary key ccaa); the variables of blocks E and F are exclusive to the main layer.

For each field the real level of the variable is indicated - territorial (varies by CCAA), national (same value for the 18 CCAA within the same pathology) or derived (calculated in the pipeline itself) - because that is the distinction that conditions its correct use in the modelling of section 4 (see limitation documented in point 3.5).

A.1. Identifiers and base variable

Field	Type	Description	Source	Level
anio	int	Reference year of all records (2023)	-	-
ccaa	str	Autonomous community or autonomous city (17 CCAA + Ceuta and Melilla combined into one unit)	Alta_Diagnostico_Provincia / Tablas_CCAA_2023	Territorial
patologia_cie10	str	ICD-10 chapter code, 4 digits ending in 00 (e.g. 1100)	Alta_Diagnostico_Provincia	Real
patologia_nombre	str	Descriptive name of the diagnostic chapter	Alta_Diagnostico_Provincia	Real
altas	float	Number of hospital discharges of that pathology in that CCAA (2023)	Alta_Diagnostico_Provincia	Real, territorial - base variable of the index

A.2. CCAA structural context – hospitals and bed resourcing

Field	Type	Description	Level
hospitales_sns	int	No. of public-SNS hospitals in the CCAA	Territorial
hospitales_privados	int	No. of private hospitals in the CCAA	Territorial
hospitales_total	int	Total no. of hospitals in the CCAA	Territorial
camas_instaladas_sns	int	Installed beds of the public-SNS subsector	Territorial
camas_instaladas_total	int	Total installed beds (public + private)	Territorial
camas_instaladas_por1000	float	Total installed beds per 1,000 inhabitants	Territorial
camas_funcionamiento_sns	int	Beds in operation of the public-SNS subsector	Territorial
camas_funcionamiento_total	int	Total beds in operation - denominator of altas_por1000hab_camas (A.7)	Territorial
camas_funcionamiento_por1000	float	Total beds in operation per 1,000 inhabitants	Territorial
pct_camas_sns	float	% of beds in operation belonging to the public-SNS subsector (observed range: 69.8%-100%)	Territorial

A.3. CCAA structural context - staff

Field	Type	Description	Level
personal_jornada_completa	int	Staff linked full-time	Territorial
personal_jornada_parcial	int	Staff linked part-time	Territorial
personal_vinculado	int	Total linked staff (full-time + part-time)	Territorial
personal_por1000	float	Linked staff per 1,000 inhabitants	Territorial

A.4. CCAA structural context - admission rates and hospitalisation indicators

Field	Type	Description	Level
ingresos_total_ccaa	int	Total no. of hospital admissions of the CCAA (all pathologies)	Territorial
ingresos_por1000	float	Total admissions per 1,000 inhabitants	Territorial
altas_total_ccaa	int	Total no. of discharges of the CCAA (all pathologies) - denominator of altas_pct_sobre_ccaa (A.7)	Territorial
altas_por1000	float	Total discharges per 1,000 inhabitants	Territorial
estancias_total_ccaa	int	Total no. of stay-days of the CCAA	Territorial
estancias_por1000	float	Total stays per 1,000 inhabitants	Territorial
estancia_media_sns	float	Average stay (days), public-SNS subsector	Territorial
ocupacion_sns	float	Occupancy rate, public-SNS subsector (proportion 0-1)	Territorial
rotacion_sns	float	Bed turnover rate, public-SNS subsector	Territorial
mortalidad_sns	float	In-hospital mortality, public-SNS subsector (proportion 0-1)	Territorial
estancia_media_priv	float	Average stay (days), private subsector	Territorial
ocupacion_priv	float	Occupancy rate, private subsector	Territorial
rotacion_priv	float	Turnover rate, private subsector	Territorial
mortalidad_priv	float	In-hospital mortality, private subsector	Territorial

Note: the indicators of the public-SNS subsector are taken as the reference value of the CCAA in the analysis (see point 3.3), since it concentrates between 69.8% and 100% of beds in operation depending on the territory; those of the private subsector are kept as additional variables, not as a substitute.

A.5. CCAA structural context - emergencies

Field	Type	Description	Level
urgencias_total	int	Total no. of emergencies attended in the CCAA	Territorial
pct_urgencias_sns	float	% of emergencies attended in SNS centres	Territorial
urgencias_por1000	int	Emergencies attended per 1,000 inhabitants	Territorial
pct_urgencias_ingresadas	float	% of emergencies ending in hospital admission	Territorial
presion_urgencias	float	Emergency pressure indicator of the CCAA (proportion 0-1)	Territorial

A.6. National context by pathology

Field	Type	Description	Level
tasa_traslado_nacional	float	% of discharges of that pathology whose reason was transfer to another centre (range: 1.0%-13.3%)	National - same value in the 18 CCAA for a given pathology (see 3.5)
pct_fallecimiento_nacional	float	% of discharges of that pathology whose reason was death (range: 0.004%-13.1%)	National
pct_urgente_nacional	float	% of admissions of that pathology that were urgent (range: 43.7%-80.6%)	National
estancia_media_dias_nacional	float	Average stay (days) of that pathology at national level (range: 3.0-54.3 days)	National

A.7. Derivates variables

Field	Type	Calculation	Level
altas_pct_sobre_ccaa	float	altas / altas_total_ccaa - relative weight of the pathology within its CCAA	Derived, territorial
altas_por1000hab_camass	float	altas / camas_funcionamiento_total - proxy of structural pressure of that pathology on the installed resourcing of the CCAA (not real-time occupancy)	Derived, territorial
traslados_estimados_nacional	float	altas × tasa_traslado_nacional - indicative estimate of transfers expected if the national rate of the pathology were applied to that CCAA. It is not real data on transfers by territory (it does not exist in any source): it is the national rate applied to a real territorial volume, and must be read and communicated as such	Derived, mixed (real volume × national rate)

Usage note for modelling (section 4): only the fields marked as territorial in the "Level" columns provide real variation between autonomous communities for the same pathology. The fields marked as national are constant by pathology in the 18 corresponding rows and must be treated as pathology context, never as an observed territorial difference.

Annex B. Detail of the 8 original sources and their technical sheet

The eight sources come from the Hospital Morbidity Survey (EMH; INE, 2024) and the Statistics of Health Establishments with Inpatient Care (ESCRI; Ministry of Health, 2025), both openly and publicly accessible, with no download restrictions or authorisation requirements. For each file, its technical sheet and the exact role it plays in the project (gold layer of the index, national trend block, or excluded) are indicated.

File	Body / statistical operation	Internal sheet(s)	Period	Geographic level	Role in the TFM
Alta_Diagnostico_Provincia.xlsx	INE - EMH (INE, 2024)	tabla-74901	2023	National, CCAA and province	Gold layer (section 3) - base variable of the index (discharges); only diagnosis × territory crossing available
Alta_Motivo_ingreso.xlsx	INE - EMH (INE, 2024)	tabla-74880	2023	National only	Gold layer - national context by pathology (tasa_traslado_nacional, pct_fallecimiento_nacional)
Alta_Urgencia_ingreso.xlsx	INE - EMH (INE, 2024)	tabla-74877	2023	National only	Gold layer - national context by pathology (pct_urgente_nacional)
Estancia_media.xlsx	INE - EMH (INE, 2024)	tabla-74890	2023	National only	Gold layer - national context by pathology (estancia_media_dias_nacional)
Altas_Diagnostico_principal.xlsx	INE - EMH (INE, 2024)	tabla-74878	2023	National only	Excluded from the gold layer - discharges by diagnosis × age group; redundant with the age breakdown already worked on in the EDA of section 2 (Figures 2.6.4-2.6.5), without adding a new territorial crossing
Tablas_CCAA_2023.xlsx	Ministry of Health - ESCRI (2025)	24 sheets (1.1 to 7.3, plus index and charts); sheets 1.1, 2.1, 4.1, 5.1, 5.2 and 6.4 are used	2023	CCAA	Gold layer - territorial structural context (hospitals, beds, staff, admission rates, hospitalisation indicators, emergencies)
Actividad_Evolucion_2014-2023.xlsx	Ministry of Health - ESCRI (2025)	13 sheets (TABLE 3.1 3.2 to TABLE 3.20)	2014-2023	National	Trend block (section 2) - total discharges, average stay, occupancy, emergencies
Dotacion_Evolucion_2014-2023.xlsx	Ministry of Health - ESCRI (2025)	10 sheets (TABLE 1.1 1.2 to TABLES 1.15)	2014-2023	National	Trend block (section 2) - public and private bed resourcing

Notes common to the eight sources:

- **Access type:** open and public, with no registration or payment; direct download in Excel format from the INE (ine.es) and Ministry of Health (sanidad.gob.es) websites.
- **Stability:** official sources with periodic (annual) publication, with high format stability and maintenance guaranteed by the issuing body.
- **Diagnosis coding:** ICD-10 (WHO, 2019), hierarchical (chapter/subchapter) in the same column, in the five sources that include diagnosis.
- **Statistical secrecy values:** represented as .. in low-count cells; treated as missing throughout the pipeline (section 3.2), never as zero.
- **Structural particularity:** all sheets of Tablas_CCAA_2023.xlsx, Actividad_Evolucion_2014-2023.xlsx and Dotacion_Evolucion_2014-2023.xlsx repeat the full workbook title before the real table, with a different number of header rows per sheet - hence the need for dynamic header location documented in section 3.2.

Annex C. Conceptual design not implemented conceptual no implementado

This annex preserves, with the methodological corrections already incorporated in the review of Deliverable 4, the design of the two modules discarded from the final MVP scope (section 1.4): M2 - referral pressure and M3 - 30-day readmission risk. Neither is trained, validated or implemented in this TFM; they are documented so they can be activated in the future if the real sources that do not exist today appear (see sections 6.4 and 7.3).

C.1. Why they are discarded from the MVP

After the file-by-file validation of the 8 real sources (section 2), none offers:

- A transfers panel with territorial breakdown (necessary for M2): Alta_Motivo_ingreso exists only at national level (section 3.5; INE, 2024).
- A patient or episode identifier, or a real aggregated readmission indicator (necessary for M3): none of the 8 sources contains it.

Both modules would require, to be sustained with the current sources, resorting to synthetic data - something the project explicitly excludes as a methodological principle since section 1.

C.2. M2 - Referral pressure

Approach: regression (Ridge / Gradient Boosting Regressor; Hoerl and Kennard, 1970; Friedman, 2001) on tasa_traslado, with predictors mortalidad_intrahosp_pct, desviacion_estancia, camas_totales_ccaa (normalised by the CCAA's total discharges) and anio.

Methodological correction incorporated: camas_totales_ccaa is annual structural context, and must never be read or communicated as real bed availability for recommending a specific referral. The score would be a planning prioritisation index, not an operational recommendation about an individual patient or transfer - the same communication principle already applied to the implemented care-pressure index (section 6.2).

Output data (conceptual):

Field	Description	Type
patologia_cie10, ccaa_codigo, anio	Predicted unit	str/int
score_presion_derivacion	Index from 0 to 100; higher value = higher structural referral pressure	float
factores_principales	2-3 variables with the greatest weight in the score	list[str]

Validation (conceptual): leave-one-year-out (cross-validation leaving one year out, given the low number of years available); R² and MAE metrics on tasa_traslado; baseline = historical mean of tasa_traslado by pathology, without CCAA.

Activation condition: a real file with a time series of transfers broken down by CCAA, non-existent today.

C.3. M3 - 30-day readmission risk

Approach: regression (Gradient Boosting Regressor; Friedman, 2001) on tasa_reingreso_30d, with SHAP explainability (Lundberg and Lee, 2017), predictors desviacion_estancia, mortalidad_intrahosp_pct, pct_ingreso_urgente, grupo_edad (ordinally encoded) and patologia_cie10.

Mandatory methodological corrections, incorporated after the review of Deliverable 4:

1. **Validation by real patient, not by episode.** If the design ever reached individual-episode level, splitting by episodio_id does not prevent several episodes of the same person from appearing simultaneously in training and test, which artificially inflates the metrics - exactly the same type of pseudoreplication leakage that was corrected in the evaluation of the implemented index (section 5.1, GroupKFold; Hurlbert, 1984; Roberts et al., 2017). Validation must be by groups (GroupKFold on a real patient identifier) and, preferably, also temporal.
2. **Risk of circularity with a synthetic label.** If the readmission rate were ever completed with synthetic data to gain granularity, the model could merely reproduce the rules used to generate that label, obtaining high metrics without real clinical validity (Kaufman et al., 2012). For that reason, as long as the target variable does not come from a real aggregated indicator (iCMBD), M3 is not trained - not even as a technical test without explicitly declaring that it is a validation of the prototype, never evidence of real care capacity.

An individual binary classifier is discarded, as already documented in Deliverable 3: it would require a per-patient readmission label that no public source offers without resorting to synthetic data.

Output data (conceptual):

Field	Description	Type
patologia_cie10, grupo_edad, ccaa_codigo, anio	Predicted unit	str/int
score_riesgo_reingreso	Index from 0 to 100; higher value = higher relative readmission risk in that segment	float
factores_principales	2-3 variables with the greatest weight in the score	list[str]

Intended consumption: ranking/map by CCAA and pathology, never a recommendation about an identified individual patient - the dashboard would explicitly label these panels as "segment-level prioritisation indices", the same language already applied to the implemented panel (section 6.2).

Activation condition: availability of iCMBD (or equivalent) as a real aggregated readmission indicator, with the pathology and CCAA granularity confirmed.

C.4. Specific risks of M2/M3 (conceptual)

Risk	Description	Contingency plan
Circularity with a synthetic label (M3)	The model could reproduce the generation rules of a synthetic label instead of demonstrating clinical validity (Kaufman et al., 2012)	M3 is not trained while the target variable does not come from iCMBD; if explored with synthetic data for technical reasons, the result would be presented as technical validation of the prototype, never as real care evidence
Confusion between clinical severity and network capacity (M2)	The model could attribute to "pathology complexity" what is really a lack of beds in a specific CCAA	Same risk already identified and explicitly treated in the implemented index (section 5.5)
Information leakage between episodes of the same patient (M3)	Splitting by episodio_id does not prevent two episodes of the same person from falling one in training and one in test (Roberts et al., 2017)	Validation by groups on a real patient identifier, combined with temporal separation
Real granularity of iCMBD	The readmission indicator may not break down by pathology and CCAA simultaneously	Simplify to the breakdown the indicator does offer, documenting the loss of resolution
Low volume in pathology × CCAA × age combinations	Unstable rates, overfitting risk	Minimum threshold of discharges per cell; grouping of infrequent pathologies - same type of risk already observed in the implemented index (section 5.5, Ceuta and Melilla and La Rioja)

C.5. General activation condition

Both modules would only be activated after confirming the real granularity of the sources that are not available today (iCMBD for M3, territorial transfers for M2), and with the MVP (care-pressure index, sections 3 to 6) already delivered and evaluated - applying from the outset the group-validation corrections, never by simple episode, set out in C.3.